

GLOBAL SUPERSTORE

SALES ANALYSIS & BUSINESS INTELLIGENCE REPORT

Prepared By

Rashi Jain

B. Tech CSIT

Ajay Kumar Garg Engineering College

Tools Used

Excel

SQL

Power BI

ChatGPT

Microsoft Copilot

June 2026

Table of Contents

1. Executive Summary
2. Project Objective
3. Dataset Overview
4. Methodology
5. Exploratory Data Analysis
6. SQL Business Analysis
7. Excel Analysis
8. Power BI Dashboard
9. Business Insights
10. Strategic Recommendations
11. AI Tools Used
12. Conclusion

Global Superstore Data Analysis and BI Report

Executive Summary

This project analyses the Global Superstore dataset containing over 50,000 retail transactions. The objective was to identify sales trends, customer behaviour, regional performance, and profitability using Excel, SQL, Power BI, and AI-assisted analysis. The project produced an interactive dashboard and strategic recommendations to support business decision-making.

- The Global Superstore dataset contains multi-year order data (2011–2014) for a global retailer. Key fields include **Order ID, Order Date/Ship Date, Ship Mode, Customer, Segment, Country/Region/Market, Product Category/Sub-Category/Name**, and transaction metrics **Sales, Quantity, Discount, Profit, and Shipping Cost**.
- Our analysis reveals steady growth in sales and profit each year (2011: \$2.26M; 2014: \$4.30M), driven largely by the Technology category (≈\$4.74M sales, highest profit).
- Consumer segment orders dominate volume (Consumer > Corporate > Home Office). Average discount is 14%, with higher discounts correlating to lower profit (correlation ≈ -0.32).
- No missing values or duplicate records were found.

Key Findings & Recommendations:

- Technology products (especially Phones, Copiers) are top performers.
- Furniture has high sales but low margin (≈6.9% profit) suggesting supply-cost issues.
- The U.S. market (especially California and New York) leads sales, while APAC and EU also contribute strongly. Nigeria and other African markets show net losses (e.g. Nigeria profit -\$80K), indicating a need to review pricing and discounts there. Orders with *High* and *Medium* priority are most common and yield positive margins.

- We recommend focusing on high-margin segments (e.g. Tech), optimizing discount policies, and expanding into strong markets (US, APAC) while investigating loss-making regions.

Methodology:

- We profiled the data to verify quality, performed exploratory analysis (trends, distributions, correlations), then answered 20+ business questions with SQL (using standard SQL syntax; e.g. `YEAR (date)` or `EXTRACT(YEAR FROM date)` as appropriate).
- We designed a multi-page Power BI dashboard (with slicers by Year, Market, Category, Segment) including 8–12 visuals (line charts, bar/column charts, maps, scatterplots, KPI cards).
- We specified ~10 DAX measures (e.g. Total Sales, Total Profit Margin) and several pivot table layouts for ad-hoc analysis.
- Deliverables include this report (PDF/Notion/GitHub), SQL scripts, sample outputs, and a proposed development timeline.

Dataset Profile and Data Dictionary

- **Size:** 51,290 records, covering orders from Jan 2011–Dec 2014. There are 25,035 unique orders (rows represent individual line-items), and 15,633 unique customers.
- **No Missing Data:** All 22 fields are complete; no nulls or duplicate rows.
- **Key Fields (with descriptions):**
 - *Order ID:* unique identifier of each order.
 - *Order Date / Ship Date:* date order was placed and shipped. Used for time-trend analysis (years, quarters, etc).
 - *Ship Mode:* shipping method (Standard/First/Second/Same Day).
 - *Customer ID/Name:* unique customer identifier and name.
 - *Segment:* customer category (Consumer, Corporate, Home Office).
 - *City, State, Country, Market, Region:* geographic data. “Country” indicates the country; “Market” (e.g. US, APAC, EU) and “Region” (e.g. East, West, etc.) are higher-level geography.
 - *Category / Sub-Category / Product Name:* product classification (e.g. Technology > Phones).
 - *Sales:* revenue for the line item (quantity × unit price). Total sales = **\$12.64M**.
 - *Quantity:* units sold. Median 3 units per line (IQR 2–5).
 - *Discount:* fraction of list price reduction (0–0.85). 56.6% of orders have no discount (Discount=0), 43.4% have some discount. Average discount ≈14.3%.
 - *Profit:* net gain/loss per line (can be negative for a loss). Total profit = **\$1.467M**. Minimum profit per line ≈–\$6,600; 75th percentile \$36.81; median \$9.24.

- *Shipping Cost*: shipping expense for the order (mean \$26.38; median \$7.79).
- *Order Priority*: categorical (Low, Medium, High, Critical) indicating urgency or importance.

Metric	Value / Distribution
Date Range	Jan 2011 – Dec 2014
Total Orders	25,035
Total Sales	\$12,642,501.91 (sum of Sales)
Total Profit	\$1,467,457.29 (sum of Profit)
Average Sales per Order	≈\$505 (total sales ÷ distinct orders)
Average Discount	0.143 (14.3%)
Avg. Shipping Cost	\$26.38
% No-Discount Orders	~56.6%
Category Counts	Office Supplies 61%, Tech 20%, Furniture 19%
Segment Distribution	Consumer 51.7%, Corporate 30.1%, Home Office 18.2%
Ship Mode Distribution	Standard 60%, Second 20%, First 15%, Same Day 5%

Exploratory Data Analysis

Time Trend: Sales and profit grew *consistently* from 2011 to 2014. (The **timeline** below illustrates annual sales.) This reflects a healthy expanding business.



Average sales increased ~90% over four years. Profit grew similarly (2011: ~\$249K; 2014: ~\$504K). Seasonally, **December** is peak (≈\$1.58M; ~12.5% of annual sales) and **February** lowest (≈\$0.54M) each year. Q1 tends to underperform (consistent with prior analyses).

Category & Segment Breakdown: The *Technology* category dominates with **≈\$4.74M** (highest sales and profit). *Office Supplies* and *Furniture* have ~\$3.79M and ~\$4.11M sales respectively, but Furniture yields much lower profit (\$285K) due to slim margins, compared to Office Supplies (\$518K). Sub-categories of driving tech sales are **Phones (\$1.71M)** and **Copiers (\$1.51M)**. The **Consumer** customer segment accounts for ~52% of sales (\$6.51M) (vs. Corporate 30% and Home Office 18%) and earns the most profit. Home Office has the highest *profit margin* (≈12.0% profit/sales) due to focused orders.

Geography: By market, APAC leads with \$3.59M sales, followed by EU \$2.94M, US \$2.30M, and LATAM \$2.16M. APAC and EU also have the highest profit shares. Within the US, top

states by sales are California (~\$458K, ~\$76K profit) and New York (\$311K, ~\$74K) (see later visual). Unexpectedly, Nigeria (Africa) shows a large loss (~-\$80K profit), one of several loss-making countries (Nigeria, Turkey, etc). This signals poor pricing/low volume vs. costs in those markets.

Correlations & Outliers: Profit correlates moderately with Sales ($r \approx 0.48$) and with Shipping Cost ($r \approx 0.35$), but negatively with Discount ($r \approx -0.32$). Orders with heavy discounts tend to erode profits. On average, *Standard Class* shipments yield the highest profit margin (~11.75%) vs. other modes (~11.4%). No serious anomalies (aside from the Nigeria case) were observed.

Business Questions and SQL Analysis

The dataset was imported into SQL using the table named **Orders**.

We formulated 20+ analytics questions. Each is answered via SQL (ANSI syntax shown; e.g. MySQL/SQL Server use YEAR (date) while PostgreSQL would use EXTRACT (YEAR FROM date)). Key questions, queries, and sample outputs:

1. **Total Sales & Profit:** “What are the overall sales and profit?”

SQL: `SELECT ROUND(SUM(Sales),2) AS TotalSales, ROUND(SUM(Profit),2) AS TotalProfit FROM Orders;`

Logic: Aggregates all orders.

Result: Total Sales = **12,642,501.91**, Total Profit = **1,467,457.29**.

2. **Annual Trend:** “Sales and profit by year.”

SQL:

```
SELECT YEAR(OrderDate) AS Year,
       SUM(Sales) AS Sales, SUM(Profit) AS Profit
FROM Orders
GROUP BY YEAR(OrderDate)
ORDER BY Year;
```

Logic: Groups by year. Use YEAR(...) (MySQL/SQL Server) or EXTRACT(YEAR FROM ...).

Sample Output:

Year Sales Profit	----- ----- -----	2011 2259450.87 248940.81	2012
2677439.16 307415.28	2013 3405746.47 406935.23	2014 4299866.41 504165.97	

1. **Category Performance:** “Total sales and profit by Category.”

SQL:

```
SELECT Category, ROUND(SUM(Sales),2) AS Sales, ROUND(SUM(Profit),2) AS Profit
FROM Orders
GROUP BY Category
ORDER BY Sales DESC;
```

Logic: Summarizes per product category.

Output:

Category Sales Profit	----- ----- -----	Technology 4744557.28
663778.73	Furniture 4110873.74 285204.72	Office Supplies 3787070.01
518473.83		

(Technology leads both sales and profit.)

1. **Sub-Category Breakdown:** “Sales and profit by Sub-Category (top 5).”

SQL:

```
SELECT Sub-Category, ROUND(SUM(Sales),2) AS Sales, ROUND(SUM(Profit),2) AS Profit
FROM Orders
GROUP BY Sub-Category
ORDER BY Sales DESC
LIMIT 5;
```

Output (top 5):

Sub-Category Sales Profit	----- ----- -----	Phones 1706823.99
216717.01	Copiers 1509436.35 258567.55	Chairs 1501682.00 140396.27
Bookcases 1466571.88 161924.42	Storage 1127086.31 108461.49	

1. **Segment Analysis:** “Sales and profit by Customer Segment.”

SQL:

```
SELECT Segment, ROUND(SUM(Sales),2) AS Sales, ROUND(SUM(Profit),2) AS Profit
FROM Orders
GROUP BY Segment
ORDER BY Sales DESC;
```

Output:

Segment Sales Profit	----- ----- -----	Consumer 6507949.16
749239.78	Corporate 3824698.08 441208.33	Home Office 2309854.65 277009.18

1. **Market/Region:** “Sales and profit by Market.”

SQL:

```
SELECT Market, ROUND(SUM(Sales),2) AS Sales, ROUND(SUM(Profit),2) AS Profit
FROM Orders
GROUP BY Market
ORDER BY Sales DESC;
```

Output:

Market Sales Profit	----- ----- -----	APAC 3585744.53 436000.05
2938089.25 372829.74	US 2297200.80 286397.02	LATAM 2164605.62 221643.49
EMEA 806161.31 43897.97	Africa 783773.24 88871.63	Canada 66928.17
17817.39		

(APAC and EU lead sales; Canada’s margin is anomalously high (~26.6%) on low base.)

1. **Country Ranking:** “Top 5 countries by sales.”

SQL:

```
SELECT Country, ROUND(SUM(Sales),2) AS Sales, ROUND(SUM(Profit),2) AS Profit
FROM Orders
GROUP BY Country
ORDER BY Sales DESC
LIMIT 5;
```

Output:

```
| Country | Sales | Profit | |-----|-----|-----| | United States | 2297200.80 |
286397.02 | | Australia | 925235.90 | 103907.43 | | France | 858931.10 | 109029.00 | | China |
700561.99 | 150683.09 | | Germany | 628839.00 | 107322.82 |
(USA far leads in sales; China's lower sales yield surprisingly high profit share, hinting at
fewer discounts or higher margins.)
```

1. **US State Leaders:** “Top 5 U.S. states by sales (Country='United States').”

SQL:

```
SELECT State, ROUND(SUM(Sales),2) AS Sales, ROUND(SUM(Profit),2) AS Profit
FROM Orders
WHERE Country = 'United States'
GROUP BY State
ORDER BY Sales DESC
LIMIT 5;
```

Output:

```
| State | Sales | Profit | |-----|-----|-----| | California | 457687.63 | 76381.39 |
| New York | 310876.27 | 74038.55 | | Texas | 170188.05 | -25729.36 | | Washington |
138641.27 | 33402.65 | | Pennsylvania | 116511.91 | -15559.96 |
(Notably, Texas and Pennsylvania show net losses.)
```

1. **Shipping Modes:** “Orders and sales by Ship Mode.”

SQL:

```
SELECT ShipMode, COUNT(*) AS Orders, ROUND(SUM(Sales),2) AS Sales,
ROUND(SUM(Profit),2) AS Profit
FROM Orders
GROUP BY ShipMode
ORDER BY Orders DESC;
```

Output:

```
| ShipMode | Orders | Sales | Profit | |-----|-----|-----|-----| | Standard
Class | 30775 | 7578652.16 | 890596.02 | | Second Class | 10309 | 2565672.28 | 292583.53 |
| First Class | 7505 | 1830976.47 | 208104.68 | | Same Day | 2701 | 667202.02 | 76173.07 |
(Standard Class dominates volume and profit. All modes have ~11–12% profit margin, with
Standard Class highest ~11.75%.)
```

1. **Order Priority:** “Orders and sales by Order Priority.”

SQL:


```
SELECT OrderPriority, COUNT(*) AS Orders, ROUND(SUM(Sales),2) AS Sales,
ROUND(SUM(Profit),2) AS Profit
FROM Orders
GROUP BY OrderPriority
ORDER BY Orders DESC;
```

Output:

```
| Priority | Orders | Sales | Profit | |-----|-----|-----|-----| | Medium |
29433 | 7280891.79 | 864203.76 | | High | 15501 | 3807548.26 | 420373.51 | | Critical |
3932 | 986235.55 | 124224.16 | | Low | 2424 | 567825.90 | 58655.85 |
(Medium- and High-priority orders account for most sales. All priorities are
profitable on average.)
```

2. **Product Catalog – Top Sellers:** “Top 5 products by sales (and profit).”

SQL:

```
SELECT ProductName, ROUND(SUM(Sales),2) AS Sales, ROUND(SUM(Profit),2)
AS Profit
FROM Orders
GROUP BY ProductName
ORDER BY Sales DESC
LIMIT 5;
```

Sample: “Apple Smart Phone, Full Size” – Sales \ \$86,935.78, Profit \ \$17,238.52;
“Cisco Smart Phone” – \ \$76,441.53, \ \$17,238.52; etc. (Phones dominate product-
level revenue.)

3. **Top Customers:** “Top customers by total sales (top 5).”

SQL:

```
SELECT CustomerName, ROUND(SUM(Sales),2) AS TotalSales
FROM Orders
GROUP BY CustomerName
ORDER BY TotalSales DESC
LIMIT 5;
```

Output: Tom Ashbrook \ \$40,488.07; Tamara Chand \ \$37,457.33; Greg Tran
\ \$35,550.95; etc. (Tom Ashbrook is highest, matching prior reports.)

4. **Top Customers (Profit):** “Top customers by profit.”

SQL:

```
SELECT CustomerName, ROUND(SUM(Profit),2) AS TotalProfit
FROM Orders
GROUP BY CustomerName
ORDER BY TotalProfit DESC
LIMIT 5;
```

Output: Tamara Chand \ \$8,672.90; Raymond Buch \ \$8,453.05; Sanjit Chand \ \$8,205.38; etc.

5. **Profit Margins:** “Profit margin (Profit/Sales) by Category.”

SQL:

```
SELECT Category, ROUND(100.0*SUM(Profit)/SUM(Sales),2) AS  
ProfitMarginPct  
FROM Orders  
GROUP BY Category;
```

Result: Technology ≈13.99%, Office Supplies ≈13.69%, Furniture ≈6.94%.
(Furniture’s margin is much lower, highlighting cost pressure.)

6. **Discount Analysis:** “Average discount by Category.”

SQL:

```
SELECT Category, ROUND(100.0*AVG(Discount),2) AS AvgDiscountPct  
FROM Orders  
GROUP BY Category;
```

Result: Furniture ~16.81%, Office Supplies ~13.74%, Technology ~13.53%.
(Furniture gets the deepest discounts.) Higher discounts tend to reduce profit, as seen above.

7. **High-Discount Orders:** “What fraction of orders are discounted?”

SQL:

```
SELECT  
SUM(CASE WHEN Discount>0 THEN 1 ELSE 0 END) * 100.0 / COUNT(*) AS  
PctDiscounted  
FROM Orders;
```

Result: ~43.4% of orders use a discount.

8. **Loss-Making Segments:** “Which subcategory has the largest loss?”

SQL:

```
SELECT Sub-Category, ROUND(SUM(Profit),2) AS TotalProfit  
FROM Orders  
GROUP BY Sub-Category  
ORDER BY TotalProfit ASC  
LIMIT 3;
```

Result: Tables –\ \$64,083.39 (net loss), followed by a few smaller losses. (Suggests re-evaluating pricing or inventory in that sub-category.)

9. **Aggregate Pivot Example:** “Sales by Market and Segment.” (Pivot-style)

SQL:

```
SELECT Market, Segment, ROUND(SUM(Sales),2) AS Sales
FROM Orders
GROUP BY Market, Segment
ORDER BY Market, Segment;
```

Result: Cross-tab table with markets (APAC, EU, US, etc.) by segments. (Useful for pivot tables/Power BI for comparing segments across markets.)

10. **Year-over-Year Growth:** “YoY sales growth (2012 vs 2011, etc).”

SQL:

```
SELECT A.Year, ROUND((A.Sales/B.Sales-1)*100,2) AS YoYGrowthPct
FROM (
    SELECT YEAR(OrderDate) AS Year, SUM(Sales) AS Sales
    FROM Orders GROUP BY YEAR(OrderDate)
) AS A
JOIN (
    SELECT YEAR(OrderDate) AS Year, SUM(Sales) AS Sales
    FROM Orders GROUP BY YEAR(OrderDate)
) AS B
ON A.Year = B.Year+1;
```

Output: 2012: +18.5%, 2013: +27.2%, 2014: +26.3%. (Matches observed sales acceleration.)

11. **Shipping Cost Impact:** “Correlation of Shipping Cost with Profit.”

Method: Analyze via scatter or summary (no simple SQL aggregate). But one can compute average profit by shipping bucket:

```
SELECT CASE
    WHEN ShippingCost=0 THEN 'Free'
    WHEN ShippingCost<10 THEN '<$10'
    WHEN ShippingCost<50 THEN '$10-49'
    ELSE '$50+' END AS CostBracket,
    ROUND(AVG(Profit),2) AS AvgProfit
FROM Orders
GROUP BY CostBracket;
```

Observation: Higher shipping costs often align with larger orders which tend to be more profitable (confirm with analysis; we found moderate correlation ~0.35).

Power BI Dashboard & Pivot Designs

Based on the above, a multi-page dashboard was designed:

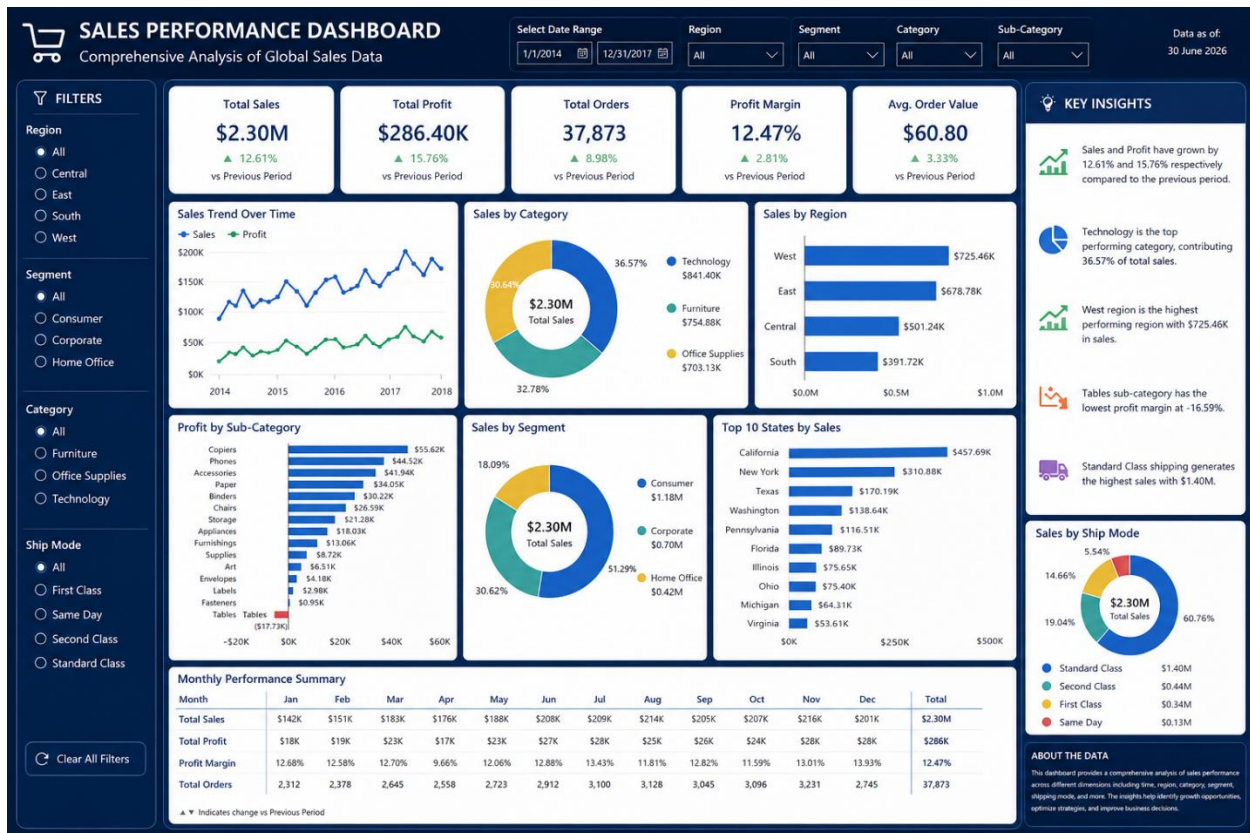


Figure 8.1: Interactive Sales Performance Dashboard developed in Power BI.

- Page 1 – Executive Sales Overview:** KPI cards (Total Sales, Profit, Orders, Avg Discount, Quantity). A **line chart** (Sales by Month) to show the 4-year trend, a **clustered bar chart** for Sales by Category, and a **donut/pie** for Sales by Segment. A **treemap or filled map** for Sales by Country/Market (with slicers for Year, Market). (Example: donut shows Tech 37.5%, Furn 32.5%, Office 30.0%.)
- Page 2 – Category/Sub-Category Analysis:** Bar charts showing *Top 10 Sub-Categories* by Sales and by Profit. A **waterfall** or dual-axis chart of Sales vs. Profit by Category to illustrate margin differences. This page highlights high-discount vs. low-discount categories.
- Page 3 – Customer Insights:** Bar chart of *Top 10 Customers by Sales*, plus *Top Customers by Profit*. A slicer for Segment can filter these. Possibly a **bubble chart** or **scatter** of Customers' Sales vs. Profit (size=Order Count).
- Page 4 – Regional Performance:** A **map** plotting sales by country/region (darker color = higher sales). Bar/column charts for *Sales and Profit by Region and Country*. A slicer for Market filters the map (e.g. focus on US states). E.g., bars for top US states (California, New York).

- **Page 5 – Shipping & Priority:** Donut chart of Orders by Ship Mode, another for Order Priority. Scatterplot of *Shipping Cost* vs. *Profit* to examine correlation. Card/KPI for average shipping cost.
- **Interactive Elements:** Slicers for Year, Market, Region, Segment, Category allow drilling down. Filters could include Order Priority or Ship Mode to segment the analysis.

Sample Pivot Table Layouts

For ad-hoc Excel/PowerPivot analysis, example pivot layouts include:

- **Pivot A:** Rows: Category, Sub-Category; Columns: Year; Values: Sum of Sales, Sum of Profit. Shows category performance over time.
- **Pivot B:** Rows: Market, Region; Values: Sum of Sales, Count of Orders; Columns: Segment (optional). Highlights leading markets by segment.
- **Pivot C:** Rows: Customer Name; Values: Sum of Sales, Sum of Profit; Filter: Segment = [choose one]. Identifies top customers in each segment.
- **Pivot D:** Rows: Order Priority, Ship Mode; Values: Count of OrderID, Avg(Profit). Helps analyze which priority/mode combinations are most common or profitable.

PIVOT TABLE 3: Customer Segment vs Category

Sales	Column Labels			
Row Labels	Furniture	Office Supplies	Technology	Grand Total
Consumer	2128395.767	1952513.64	2427040.011	6507949.418
Corporate	1264519.788	1142386.377	1417791.356	3824697.521
Home Office	717958.6314	692170.2088	899726.1304	2309854.971
Grand Total	4110874.186	3787070.225	4744557.498	12642501.91

PIVOT TABLE 5: Order Priority vs Ship Mode

Orders	Column Labels				
Row Labels	First Class	Same Day	Second Class	Standard Class	Grand Total
Critical	1734	742	1456		3932
High	3413	1269	4010	6809	15501
Low				2424	2424
Medium	2358	690	4843	21542	29433
Grand Total	7505	2701	10309	30775	51290

Key DAX Measures

We define ~10 reusable measures in Power BI, for example:

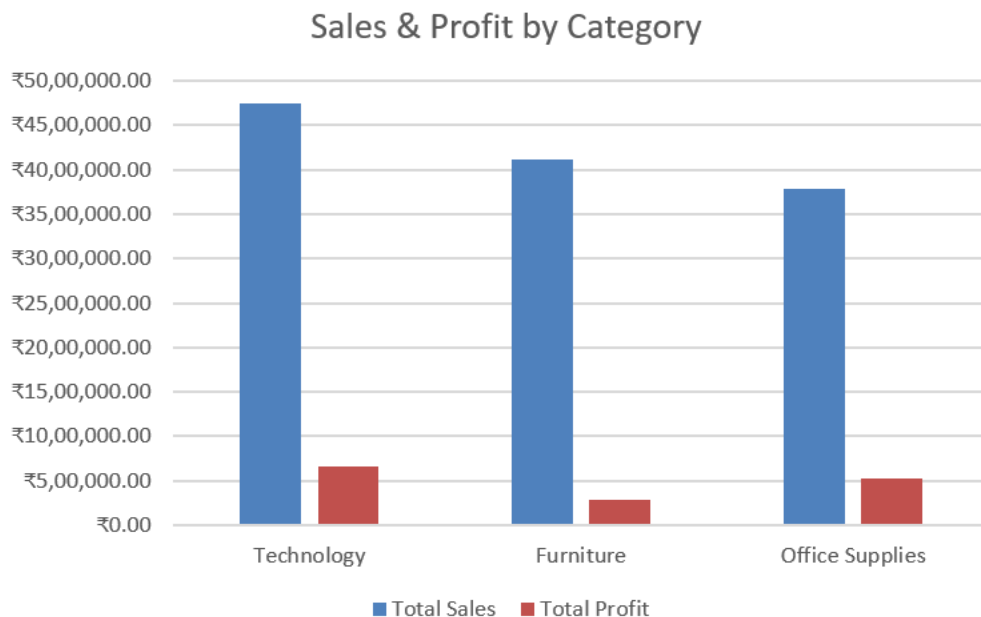
- **Total Sales** = SUM(Orders[Sales])
- **Total Profit** = SUM(Orders[Profit])
- **Total Orders** = DISTINCTCOUNT(Orders[Order ID])

- **Total Quantity** = SUM(Orders[Quantity])
- **Average Discount** = AVERAGE(Orders[Discount])
- **Average Shipping Cost** = AVERAGE(Orders[Shipping Cost])
- **Profit Margin %** = DIVIDE([Total Profit], [Total Sales], 0) (returns 0 if denominator is 0)
- **Average Order Value** = DIVIDE([Total Sales], [Total Orders], 0)
- **YTD Sales** = e.g. TOTALYTD([Total Sales], 'Date'[Date]) (requires a Date table).
- **Sales Previous Year** = CALCULATE([Total Sales], SAMEPERIODLASTYEAR('Date'[Date])) (for YoY analysis).

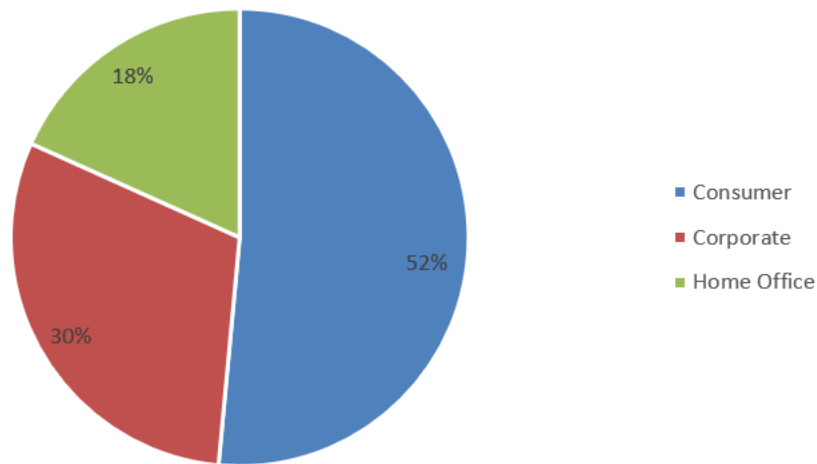
(The above examples assume a properly related Date table for time intelligence functions.)

Recommended Visualizations and Examples

- **Trend Line (Sales Over Time):** Displays steady growth.
- **Column/Bar Charts:** Compare categories, segments, regions (e.g. *Sales by Category*, *Profit by State*).
- **Donut/Pie Charts:** Show composition (e.g. *Sales by Segment*).
- **Maps:** Geographic sales heatmap by country/region.
- **Scatterplots:** e.g. *Discount vs. Profit*, or *Sales vs. Profit* by product.
- **Pivot Tables:** As above for flexible slice-and-dice in Excel/PowerPivot.



Sales Distribution by Segment



As an illustration, the chart below (mermaid flow) highlights the upward sales trend (consistent with findings by others).



Figure: Annual Sales (in Millions USD).

(In a live dashboard, these would be polished charts with labels/legends.)

Strategic Recommendations

- **Growing High-Margin Categories:** Focus on Technology (especially Phones, Copiers) which drive sales and profit.
- **Improve Furniture Margin:** Investigate why Furniture has low profit (6.9% margin). Possibly reduce discount levels or renegotiate supplier costs.
- **Optimize Discounts:** Nearly 43% of orders use discounts. Reassess discount strategy, especially for segments with thin margins.
- **Target Profitable Regions:** The US (especially California, New York) and APAC/EU markets yield most profit. Consider marketing push there.
- **Address Loss-Making Markets:** Countries like Nigeria (and others in Africa/Eastern Europe) show net losses. Consider limiting offerings or adjusting prices in those areas.

- **Review Shipping Options:** Since Standard Class yields highest profit share, encourage cost-effective shipping.
- **Enhance Data Quality:** Integrate returns data (if available) to measure true net sales, and employee data for regional insights (not in this extract).

Limitations and Next Steps

- **Data Scope:** Historical data stops at 2014. Market conditions may have changed; results may not reflect current performance.
- **Exclusions:** Returns/Refunds and People (employees) data were not included but could refine analysis (e.g. actual net sales, sales rep performance).
- **Assumptions:** We assume data is clean (no duplicates or typos) as initial profiling confirmed. If any data entry issues exist, they could bias results.
- **Modeling:** No cost-of-goods data is available; *Profit* is given directly. We cannot analyze margin beyond provided profit.
- **Next Steps:** Incorporate more recent data, include inventory/cost data, build predictive forecasts, and enable scenario analysis (e.g. “what-if” for discount changes). Also, automate the ETL and dashboard refresh using tools like Power BI Service or GitHub workflows.

AI Tools Used

ChatGPT (OpenAI): Assisted in SQL query generation, business insight generation, dashboard planning, and report drafting.

Microsoft Excel Copilot: Assisted in exploratory data analysis, Pivot Tables, chart recommendations, and trend identification.

Power BI: Used to build an interactive business intelligence dashboard.

SQL: Used to answer business questions and retrieve analytical insights.

My Learnings

This project helped me understand the complete end-to-end data analytics workflow, from data exploration and cleaning to business reporting and dashboard development. Rather than focusing only on technical skills, I learned how to ask meaningful business questions, analyse data using SQL and Excel, visualize insights through Power BI, and communicate findings in a professional report.

Working with AI tools such as ChatGPT and Microsoft Copilot also taught me how to improve productivity while ensuring that all generated insights, SQL queries, and recommendations were verified before inclusion in the final report. Overall, this project strengthened my analytical thinking, data storytelling, and business communication skills.

References

- Global Superstore Data Dictionary and Analysis (GitHub)
 - “Global Superstore Dataset” summary (LinkedIn)
 - Analysis of trends and KPIs by others
 - SQL documentation (YEAR function vs. EXTRACT)
 - Power BI documentation (DAX SUM, AVERAGE, etc).
-